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# Does customer concentration affect corporate risk-taking? Evidence from China

Xiyang Cao, Jian Ni, Fei Wang, Yue Xu *

*School of Finance, Southwestern University of Finance and Economics, Chengdu 611130, China*

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ABSTRACT

This paper investigates the impact of customer concentration on corporate risk-taking in China. Our analysis shows that customer concentration increases major customers' bargaining power, allowing them to transfer some of their risk to upstream firms, resulting in upstream firms' passive risk-taking. Due to their relatively weak bargaining position, firms with a concentrated customer base are forced to maintain high levels of inventories and receivables. Unlike previous research, which has frequently emphasized the positive aspects of risk-taking, our findings highlight the negative aspects of risk-taking caused by customer concentration as a result of supply chain interactions.

## 1. Introduction

Customer concentration has received much attention in recent years (e.g., Dong et al., 2021; Lee et al., 2020; Saboo et al., 2017). Many studies have shown that customer concentration increases the bargaining power of customer firms (e.g., Crawford et al., 2020; Saboo et al., 2017; Kwak and Kim, 2020; Patatoukas, 2012). In particular, Campello and Gao (2017) have found that a higher customer concentration leads to a higher cost of borrowing, and a plausible reason for this result is that customer concentration may drive upstream firms to undertake riskier investments, and this passive risk-taking of the firm due to supply chain interaction in turn raises the credit risk and inflates the cost of borrowing.

Thus, there is a concern that firms may take risky yet inferior investments (passive risk-taking) due to the pressure from concentrated customers who have relatively strong bargaining position in supply chain interactions. However, there is still little research on the relationship between customer concentration and passive risk-taking due to supply chain interactions. In fact, the existing research on corporate risk-taking tends to emphasize the positive side of risk-taking of firms, that is, entrepreneurs actively select risky projects with positive net present value to enhance firm value by leveraging their professional skills and information advantages. Many empirical papers in this research line have investigated the main causes and potential benefits of corporate risk-taking from a variety of perspectives, such as macroeconomics (Bruno and Shin, 2014), corporate governance (Habib and Hasan, 2017; John et al., 2008), and capital markets (Rajgopal and Shevlin, 2002). But in practice, a firm's choice of risky investment projects can often be influenced by its major customers. Existing research has seldom analyzed how the strategic interaction with major customers should affect the risk-taking strategy of the firm.

This study therefore empirically analyzes the relationship between customer concentration and risk-taking using data of listed firms in China, where firms often rely heavily on major customers to obtain financing resources. In fact, China has experienced rapid

* Corresponding author.
*E-mail address:* xuyue_2007@163.com (Y. Xu).

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economic growth over the past few decades. With the economic progress, there is a substantial development of the non-state-owned firms (non-SOEs) and the associated risk-taking behaviors of entrepreneurs aiming to take risky investments that can foster firm growth (Farag and Mallin, 2018; Chen and Huang, 2023). But not all the risk-taking behaviors are economically efficient in China. In light of the generally-weak bargaining position of relatively low-tech, labor intensive industries in China, some risk-taking behaviors of Chinese firms may actually cause problems such as excess capacity and inventory, and thus have negative impacts on firm value (Fulco, 2016). Although China has become the world's largest exporter and producer of industrial goods (Huang and Fan, 2022), there has been an oversupply of manufacturing products in recent years. Suppliers in China are often more susceptible to a concentrated customer base due to the intensive competition among suppliers, as well as the diversified sourcing strategies used in recent years by buyers attempting to add diversified suppliers to their supply base (Brown, 2021). As a result, we can expect a pronounced negative impact of customer concentration on the risk-taking of firms in China, which deserves empirical investigation.

To conduct the empirical analysis, we collect data on listed firms in China from 2008 to 2019 from the China Stock Market Accounting Research (CSMAR) database. We first perform a baseline regression to examine the impact of customer concentration on risk-taking, and then employ an instrumental variable method to establish causality and mitigate the potential endogeneity concerns. Our analysis shows that in the strategic supply chain interaction, upstream firms in a relatively weak bargaining position may have to make some risky investments (e.g., excessive inventory build-up) under pressure from major customers, and even more so in the presence of a concentrated customer base.

Our empirical results are relevant to developing countries where firms rely heavily on major customers. In particular, by dividing the sample into SOEs and non-SOEs, our results show that for SOEs in China, there can be significant passive risk-taking caused by concentrated customers. This result also implies the potential passive risk-taking of SOEs driven by concentrated customers in developing countries with many SOEs, such as Brazil, Vietnam and India (Nguyen and Dijk, 2012; Hennart et al., 2017; Panicker et al., 2022); indeed, since SOEs in these countries are often relatively weaker than in China, they should be more susceptible to customer concentration. Moreover, we find that the impact of customer concentration on passive risk-taking due to supply chain interaction is more pronounced for non-SOEs. Similar to the situation in China, in most developing countries firms rely heavily on supply chain partners as a source of informal financing, due to the significantly less-developed law, finance, corporate governance, and accounting standards there (Ayyagari et al., 2010; Ni et al., 2023). Thus, our analysis suggests that it is important to pay attention to the potential passive risk-taking of non-SOEs due to supply chain interactions in most developing countries.

This study can contribute to the literature as follows. First, building on Campello and Gao's (2017) tentative argument that customer concentration may drive firms to passively engage in risky investments and thus increase credit risk, our study further provides empirical evidence that customer concentration does give rise to the passive risk taking of the upstream firms. This finding highlights the necessity of considering supply chain interaction in studying corporate risk-taking, an important topic largely missing in the existing literature on risk-taking (Rajgopal and Shevlin, 2002; Habib and Hasan, 2017).

Next, our analysis can provide a more comprehensive understanding of corporate risk-taking. While existing studies emphasize the role of entrepreneurship in choosing risky investments, our study further emphasizes that concentrated customers can also influence the firm's choice of risky investments via supply chain interaction. According to Ni et al. (2023), the interaction between the firm and its large customers could be either collaborative or competitive. If the interaction is mainly collaborative, then we would expect that the firm can select better investment projects with the assistance from major customers; otherwise under competitive interaction, the investments chosen by the firm should tend to be risky and inferior under the influence and pressure of large customers. Our analysis shows that upstream firms facing the pressure of concentrated customers tend to make excessive investments in inventory and offer excessive trade credit for buyers, and a one-standard deviation increase in customer concentration can roughly increase corporate risk-taking by 12% and decrease profit by 5%. The results suggest that there is passive risk-taking under the pressure of major customers, which is consistent with the "bargaining effect" in Ni et al. (2023).

## 2. Sample and methodology

### 2.1. Sample selection and data

Following the existing literature (John et al., 2008; Liu and Wu, 2023), we construct two measures of corporate risk-taking as follows:

$$Adj_ROA_{i,t} = ROA_{i,t} - \frac{1}{N}\sum_{n=1}^{N} ROA_{i,t},$$

$$SD_ROA_{i,t} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\left(Adj_ROA_{i,t} - \frac{1}{T}\sum_{t=1}^{T} Adj_ROA_{i,t}\right)^2}\ ,\ T = 3$$

$$ROA_Diff_{i,t} = Max\left(Adj_ROA_{i,t}\right) - Min\left(Adj_ROA_{i,t}\right)$$

where *ROA* is the return on total assets, and the subscripts *i* and *t* denote the supplier company and year, respectively; *Adj_ROA* is the industry-adjusted *ROA* computed as the firm-level *ROA* minus the average *ROA* of the firm's industry each year; *SD_ROA* is the standard deviation of industry-adjusted profitability (*Adj_ROA*) over the three-year overlapping periods (i.e., 2008–2010, 2009–2011,

etc.), while *ROA_Diff* is the difference between the maximum and minimum *Adj_ROA* in the three-year overlapping periods.

We then construct two measures of customer concentration as in Patatoukas (2012) and Campello and Gao (2017). Our first measure of customer concentration is the sum of the percentage of sales to the top five customers (*CustomerSales*). The second measure, *CustomerHHI*, is a Herfindahl index of sales to the supplier's top five customers, calculated as the sum of the squares of the percentage of sales to the top five customers. A high level of *CustomerSales* or *CustomerHHI* indicates that a firm relies more heavily on major customers.

We collect annual data on listed firms in China over 2008–2019 from CSMAR and Wind databases following Ni et al. (2023). We exclude firm-year observations from the financial industry, and the final sample consists of 12,251 firm-year observations on 2584 listed firms. All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers.

### 2.2. Research model

To examine the relationship between customer concentration and corporate risk-taking, we construct the baseline regression model:

$$Risk_{i,t} = \beta_0 + \beta_1 Customer\ concentration_{i,t} + \beta_2 Controls + Ind\ FE + Year\ FE + \varepsilon_{i,t} \quad (1)$$

where $i$ and $t$ represent firm and year, respectively. *Risk* denotes one of the two measures of corporate risk-taking (*SD_ROA* and *ROA_Diff*). The independent variables are customer concentration measures, as measured by *CustomerSales* and *CustomerHHI*. We also include industry-fixed effects and year-fixed effects. In addition, our regression includes a set of control variables based on previous studies of corporate risk-taking (John et al., 2008; Faccio et al., 2011). The definitions of the main variables are presented in Table 1.

## 3. Empirical results

### 3.1. Descriptive statistics

Table 2 presents summary statistics for the main variables used in this study. The mean values of the two measures of customer concentration, *CustomerSales* and *CustomerHHI*, are 0.307 and 0.054 with corresponding standard deviations of 0.219 and 0.108, which are consistent with existing studies (Ni et al., 2023; Campello and Gao, 2017). Besides, the mean values of SD_ROA and ROA_Diff are 0.054 and 0.099, respectively, and their median values are 0.021 and 0.039, respectively; these numbers are consistent with previous studies (e.g., Huang and Fan, 2022; Faccio et al., 2016; Tian et al., 2022).

### 3.2. Baseline results

The regression results for estimating Eq. (1) are shown in Table 3. We have included both industry and year fixed effects. In all specifications, the estimated coefficients on the measures of customer concentration are positive and statistically significant (at the 1-percent level). Besides, a one-standard deviation increase in customer concentration (measured by CustomerSales) can increase SD_ROA by 12.96% from the average level, and increase ROA_Diff by 12.12%. The results suggest that the effects of customer concentration on corporate risk-taking are economically significant. The baseline results are consistent with our idea that customer

**Table 1**
Definitions of the main variables.

| Variable | Definition |
|---|---|
| ***Measures of corporate risk-taking*** | |
| *SD_ROA* | The standard deviation of rolling three-year period industry-adjusted ROA |
| *ROA_Diff* | The range of three-year rolling average of industry-adjusted ROA |
| ***Measures of customer concentration*** | |
| *CustomerSales* | The sum of the percentage of sales to the top five customers |
| *CustomerHHI* | The sum of squares of the percentage of sales for the top five customers |
| ***Control variables*** | |
| *Size* | Natural logarithm of year-end total assets |
| *Age* | Natural logarithm of one plus the number of listing years |
| *ROA* | Return on assets, calculated as net income divided by total assets |
| *TobinQ* | The ratio of market value to total assets |
| *Lev* | The ratio of total debt to total assets |
| *Largest* | The percentage of shareholding of the largest shareholder |
| *Growth* | The growth rate of sales |
| *PPE* | The ratio of fixed assets to total assets |
| ***Other variables*** | |
| *Inventory* | The ratio of inventory to sales |
| *ARTurn* | Turnover of receivables, calculated as sales divided by receivables |
| *Profit* | The operating gross margin |

**Table 2**
Summary statistics.

| Variables | *Mean* | *Std* | *Min* | *p25* | *p50* | *p75* | *Max* | *N* |
|---|---|---|---|---|---|---|---|---|
| *SD_ROA* | 0.054 | 0.086 | 0.002 | 0.010 | 0.021 | 0.047 | 0.443 | 12,251 |
| *ROA_Diff* | 0.099 | 0.156 | 0.003 | 0.020 | 0.039 | 0.089 | 0.797 | 12,251 |
| *CustomerSales* | 0.307 | 0.219 | 0.013 | 0.143 | 0.247 | 0.421 | 0.981 | 12,251 |
| *CustomerHHI* | 0.054 | 0.108 | 0.000 | 0.005 | 0.015 | 0.050 | 0.688 | 12,251 |
| *Size* | 21.913 | 1.226 | 19.357 | 21.042 | 21.761 | 22.606 | 25.652 | 12,251 |
| *Age* | 2.061 | 0.772 | 0.000 | 1.609 | 2.197 | 2.773 | 3.135 | 12,251 |
| *ROA* | 0.038 | 0.054 | -0.184 | 0.013 | 0.036 | 0.065 | 0.197 | 12,251 |
| *TobinQ* | 2.782 | 1.999 | 0.900 | 1.477 | 2.126 | 3.324 | 11.398 | 12,251 |
| *Lev* | 0.437 | 0.219 | 0.056 | 0.261 | 0.430 | 0.605 | 0.954 | 12,251 |
| *Largest* | 0.353 | 0.151 | 0.088 | 0.232 | 0.333 | 0.458 | 0.747 | 12,251 |
| *Growth* | 0.200 | 0.509 | -0.569 | -0.033 | 0.112 | 0.283 | 3.353 | 12,251 |
| *PPE* | 0.231 | 0.166 | 0.003 | 0.101 | 0.198 | 0.329 | 0.715 | 12,251 |
| *Inventory* | 0.399 | 0.703 | 0.000 | 0.109 | 0.198 | 0.360 | 4.669 | 12,227 |
| *ARTurn* | 19.49 | 33.56 | 1.422 | 3.159 | 6.081 | 15.61 | 137.3 | 12,170 |
| *Profit* | 0.272 | 0.171 | -0.028 | 0.149 | 0.240 | 0.362 | 0.798 | 12,251 |

*Notes*: This table shows the summary statistics of the main variables. See Table 1 for variable definitions. All continuous variables are winsorized at the 1st and 99th percentiles.

**Table 3**
Risk-taking and customer concentration.

| | CustomerSales | | CustomerHHI | |
|---|---|---|---|---|
| Dep. var.: | (1) | (2) | (3) | (4) |
| Risk-taking | SD_ROA | ROA_Diff | SD_ROA | ROA_Diff |
| CustomerConcentration | 0.030*** | 0.055*** | 0.042*** | 0.075*** |
| | (5.49) | (5.48) | (3.79) | (3.74) |
| Size | -0.003** | -0.006** | -0.004*** | -0.007*** |
| | (-2.42) | (-2.50) | (-3.00) | (-3.08) |
| Age | 0.004*** | 0.008*** | 0.004*** | 0.008*** |
| | (2.75) | (2.87) | (2.60) | (2.72) |
| ROA | -0.251*** | -0.466*** | -0.259*** | -0.479*** |
| | (-9.06) | (-9.24) | (-9.30) | (-9.48) |
| TobinQ | 0.005*** | 0.009*** | 0.005*** | 0.009*** |
| | (6.77) | (7.06) | (6.83) | (7.12) |
| Lev | 0.011 | 0.023 | 0.012 | 0.023 |
| | (1.42) | (1.57) | (1.43) | (1.58) |
| Largest | -0.027*** | -0.050*** | -0.027*** | -0.050*** |
| | (-3.78) | (-3.83) | (-3.76) | (-3.81) |
| Growth | 0.004** | 0.007** | 0.004** | 0.007** |
| | (2.24) | (2.22) | (2.40) | (2.37) |
| PPE | 0.009 | 0.018 | 0.006 | 0.014 |
| | (0.98) | (1.12) | (0.73) | (0.87) |
| Constant | 0.099*** | 0.183*** | 0.122*** | 0.225*** |
| | (3.80) | (3.88) | (4.75) | (4.83) |
| Industry FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 12,251 | 12,251 | 12,251 | 12,251 |
| R-squared | 0.099 | 0.103 | 0.096 | 0.100 |

*Notes:* This table shows the relation between customer concentration and corporate risk-taking. Columns (1) and (2) report the regression results for *CustomerSales*, and columns (3) and (4) report the results for *CustomerHHI*. Relevant control variables, as well as industry- and year-fixed effects are included in all regressions. See Table 1 for variable definitions. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 0.01, 0.05, and 0.10 level, respectively.

concentration will drive upstream firms to take on more risk.

### 3.3. Endogeneity concern

There are concerns about endogeneity because the model lacks an explicit source of exogenous variation in concentration. To address endogeneity concerns, we conduct instrumental variables (IV) tests and use merger and acquisition activity in customers' industries (*CustomerM&A*) as an instrument to assess the impact of customer concentration on corporate risk-taking, as described by Campello and Gao (2017).

The results of the two-stage least squares regressions are reported in Table 4. For the first-stage estimation reported in Panel A, the coefficients on *CustomerM&A* are significantly positive at the 1% level, indicating that the customer concentration increases following

M&A activity in the customer's industry. Notably, the F-statistics from the first-stage regressions pass the weak identification test, and the Kleinberg-Paap statistics pass the associated under-identification tests. Then, consistent with the baseline regression results, Panel B shows that the regression coefficients are all significantly positive at the 1% level. The IV estimates confirm our earlier findings that an increase in customer concentration leads to a higher level of corporate risk-taking for suppliers.

### 3.4. Heterogeneity analysis

This secion investigates the heterogeneity of the correlation between customer concentration and corporate risk-taking. We focus on the ownership, market competition, and the importance of the largest customer. First, this study constructs a dummy variable *SOE*, which equals 1 for a state-owned enterprise (SOE). We run subsample regressions of Eq. (1) based on *SOE*. The results are shown in Table 5. Column (1) implies the regression results of SOE samples based on Eq. (1). Meanwhile, column (2) presents the results of non-SOEs. Second, we use a dummy variable *HHI*, which equals one if the firm's Herfindahl index is greater than the sample median and zero otherwise. A lower Herfindahl index value indicates a highly competitive product market in the firm's industry and, consequently, a weaker bargaining position of suppliers relative to their customers (Fabbri and Klapper, 2016). We run subsample regressions of Eq. (1) using the *HHI*. The results for higher and lower Herfindahl indexes are shown in columns (3) and (4), respectively. Finally, this study constructs a variable *SalesLargest*. The dummy equals one if the share of total sales typically going to the firm's largest customer is less than the sample median, and zero otherwise. A larger share of sales implies more important customers and thus less bargaining power for suppliers (Fabbri and Klapper, 2016). Columns (5) and (6) show the results for small and large sales shares of the largest customers, respectively. In general, the estimated coefficient of customer concentration is more significant for non-SOEs,

**Table 4**
IV results.

| Panel A: First-stage results for IV estimations | | | | |
|---|---|---|---|---|
| *Dep. Var.:* | *CustomerSales* | | *CustomerHHI* | |
| | (1) | (2) | (3) | (4) |
| *CustomerM&A* | 0.137*** | 0.104*** | 0.044*** | 0.028*** |
| | (11.02) | (8.37) | (6.32) | (3.93) |
| Controls | No | Yes | No | Yes |
| Industry FE | No | Yes | No | Yes |
| Year FE | No | Yes | No | Yes |
| Observations | 5244 | 5244 | 5244 | 5244 |
| First-stage F-test | 121.471 | 70.086 | 39.945 | 15.478 |
| Kleibergen-Paap LM stat | 72.630 | 52.924 | 30.775 | 14.421 |
| Panel B: Second-stage results for customer concentration | | | | |
| | *CustomerSales* | | *CustomerHHI* | |
| *Dep.* var.: | (1) | (2) | (3) | (4) |
| Risk-taking | *SD_ROA* | *ROA_Diff* | *SD_ROA* | *ROA_Diff* |
| *CustomerConcentration* | 0.134*** | 0.244*** | 0.492*** | 0.899*** |
| | (3.25) | (3.28) | (2.66) | (2.68) |
| *Size* | -0.003 | -0.005 | -0.004 | -0.007 |
| | (-1.13) | (-1.18) | (-1.50) | (-1.55) |
| *Age* | 0.004 | 0.007 | 0.001 | 0.001 |
| | (1.57) | (1.64) | (0.20) | (0.24) |
| *ROA* | -0.200*** | -0.369*** | -0.196*** | -0.361*** |
| | (-4.40) | (-4.46) | (-3.74) | (-3.79) |
| *TobinQ* | 0.004*** | 0.007*** | 0.001 | 0.002 |
| | (2.69) | (2.80) | (0.50) | (0.55) |
| *Lev* | 0.025** | 0.048** | 0.027* | 0.052** |
| | (2.10) | (2.24) | (1.91) | (2.02) |
| *Largest* | -0.018 | -0.034 | -0.024 | -0.044 |
| | (-1.51) | (-1.58) | (-1.55) | (-1.60) |
| *Growth* | -0.005** | -0.009** | -0.007* | -0.013* |
| | (-2.01) | (-2.01) | (-1.92) | (-1.91) |
| *PPE* | 0.012 | 0.024 | -0.007 | -0.012 |
| | (0.87) | (0.95) | (-0.44) | (-0.39) |
| Industry FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 5244 | 5244 | 5244 | 5244 |

Notes: This table shows the results of two-stage-least-square regressions. Panel A shows the first-stage regression results of customer concentration on downstream M&A activity (*CustomerM&A*). The independent variable is *CustomerSales* in columns (1) and (2), and *CustomerHHI* in columns (3) and (4). Panel B shows the second-stage regression results of customer concentration on downstream M&A activity (*CustomerM&A*). The dependent variables are *SD_ROA* and *ROA_Diff*. The independent variable is *CustomerSales* in columns (1) and (2), and *CustomerHHI* in columns (3) and (4). Relevant control variables, as well as industry- and year-fixed effects are included in all regressions. See Table 1 for variable definitions. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 0.01, 0.05, and 0.10 level, respectively.

**Table 5**
Heterogeneity in the impacts of customer concentration of firms.

| Dep.var.: *SD_ROA* | (1) *SOEs* | (2) *non-SOEs* | (3) *L_CI* | (4) *H_CI* | (5) *L_CC* | (6) *H_CC* |
|---|---|---|---|---|---|---|
| *CustomerSales* | 0.028* | 0.031*** | 0.012* | 0.049*** | 0.013 | 0.027*** |
| | (1.72) | (5.35) | (1.74) | (5.97) | (0.65) | (3.16) |
| *Size* | -0.001 | -0.004*** | -0.003* | -0.004** | -0.004** | -0.002 |
| | (-0.24) | (-2.69) | (-1.86) | (-2.15) | (-2.16) | (-1.32) |
| *Age* | 0.002 | 0.004** | 0.007*** | 0.002 | 0.002 | 0.007*** |
| | (0.25) | (2.46) | (3.10) | (0.91) | (1.07) | (2.83) |
| *ROA* | -0.267*** | -0.247*** | -0.228*** | -0.268*** | -0.197*** | -0.292*** |
| | (-3.99) | (-8.10) | (-5.62) | (-7.16) | (-5.36) | (-7.82) |
| *TobinQ* | 0.004 | 0.005*** | 0.005*** | 0.005*** | 0.004*** | 0.006*** |
| | (1.47) | (6.38) | (4.54) | (5.03) | (4.07) | (5.64) |
| *Lev* | -0.006 | 0.014 | 0.021* | 0.006 | 0.009 | 0.013 |
| | (-0.35) | (1.59) | (1.86) | (0.59) | (0.91) | (1.21) |
| *Largest* | -0.028 | -0.028*** | -0.029*** | -0.027*** | -0.017* | -0.039*** |
| | (-1.44) | (-3.66) | (-2.83) | (-2.73) | (-1.93) | (-3.58) |
| *Growth* | 0.001 | 0.004** | 0.003 | 0.004 | 0.005** | 0.003 |
| | (0.17) | (2.35) | (1.10) | (1.57) | (2.12) | (1.23) |
| *PPE* | 0.041 | 0.002 | 0.031** | -0.020* | 0.036*** | -0.018 |
| | (1.64) | (0.18) | (2.42) | (-1.80) | (3.02) | (-1.56) |
| *Constant* | 0.062 | 0.114*** | 0.092*** | 0.126*** | 0.111*** | 0.090** |
| | (0.90) | (3.99) | (2.59) | (3.35) | (3.21) | (2.42) |
| Industry FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 1567 | 10,682 | 6116 | 6135 | 6129 | 6121 |
| *R*-squared | 0.127 | 0.097 | 0.085 | 0.160 | 0.073 | 0.113 |

Notes: This table shows the heterogeneity in the impacts of customer concentration on corporate risk-taking. The independent variable is *CustomerSales* and the dependent variable is *SD_ROA*. Columns (1) and (2) represent the regression results of SOEs and non-SOEs, respectively. Columns (3) and (4) show the regression results when competition intensity in the firm's industry is low (*L_CI*) and competition intensity is high (*H_CI*), respectively. Columns (5) and (6) show the results for a small and large sales percentage of the largest customer (*L_CC* and *H_CC*), respectively. Relevant control variables, as well as industry- and year-fixed effects are included in all regressions. See Table 1 for variable definitions. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 0.01, 0.05, and 0.10 level, respectively.

competition-intensive firms, and firms with a larger customer base, suggesting that the impact of customer concentration on firm risk-taking is more pronounced for non-state-owned firms and firms with relatively weaker bargaining power.

### 3.5. Further analysis

The above results suggest that upstream firms with a relatively weak bargaining position may have to make some risky investments under pressure from major customers, and even more so in the presence of a concentrated customer base. We now provide further evidence on the impact of bargaining power. It is well known that large customers with high bargaining power tend to pressure dependent suppliers to make concessions in terms of lowering product prices, holding extra inventory and extending trade credit (e.g. Galbraith, 1952; Lustgarten, 1975; Porter, 1974; Gosman et al., 2004; Balakrishnan et al., 1996; Lee et al., 2018; Fabbri and Klapper, 2016). Thus, if a concentrated customer base will lead to passive risk-taking through bargaining power, then upstream firms' inventories should increase with customer concentration, while accounts receivable turnover and profits should decrease.

Therefore, we expect a positive relationship between customer concentration and inventories, and a negative relationship between customer concentration and receivables turnover and upstream profits. We empirically examined the relationship between these variables and customer concentration, and the regression results are presented in Table 6.

As can be seen from columns (1) and (4) of Table 6, when the dependent variable is *Inventory*, both measures of customer concentration attract a statistically significant positive coefficient, indicating that the inventory of upstream firms generally increases with the increase in customer concentration. Meanwhile, in the other columns, the coefficients of the two measures of customer concentration are negative and statistically significant, implying that receivables turnover and profit generally decrease with the increase in customer concentration. In particular, a one-standard deviation increase of customer concentration (measured by CustomerSales) can decrease profit by about 1.4%, which is equivalent to a 5.15% reduction from the average level. In addition, Table 6 also shows that a decrease in leverage ratio is associated with an increase in profit, which may be driven by the increase in equity of more profitable firms (Frank and Goyal, 2015). Overall, this empirical evidence is consistent with our previous predictions, suggesting that firms with a concentrated customer base take on more risk due to their relatively lower bargaining power. This type of passive risk-taking behavior could be economically-inefficient and even detrimental to their operations and profitability.

Next, we further investigate the impact of delayed payments from large buyers on the firm's real and other financial activities. To this end, we conduct regression analysis on three different samples: FULL, HIGH, and LOW, where FULL represents the full sample, HIGH indicates the subsample of firms above the median in the percentage of sales from large buyers (top five customers), and HIGH

**Table 6**
Financial consequences of customer concentration.

| | CustomerSales | | | CustomerHHI | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Dep.var.: | Inventory | ARTurn | Profit | Inventory | ARTurn | Profit |
| CC | 0.266** | -18.112*** | -0.064*** | 0.461** | -19.243*** | -0.072*** |
| | (2.09) | (-6.67) | (-5.66) | (2.03) | (-3.38) | (-3.02) |
| Size | -0.063 | 3.014*** | 0.007** | -0.069* | 3.467*** | 0.008*** |
| | (-1.61) | (3.96) | (2.48) | (-1.67) | (4.56) | (3.12) |
| Age | 0.071 | 5.310*** | -0.010*** | 0.068 | 5.415*** | -0.010*** |
| | (1.44) | (5.72) | (-2.72) | (1.40) | (5.79) | (-2.60) |
| ROA | -2.968*** | 37.693*** | 0.980*** | -3.020*** | 42.794*** | 0.997*** |
| | (-2.88) | (3.40) | (19.41) | (-2.92) | (3.80) | (19.67) |
| TobinQ | -0.030 | 1.478*** | 0.018*** | -0.030 | 1.405*** | 0.018*** |
| | (-1.24) | (4.79) | (11.60) | (-1.26) | (4.53) | (11.35) |
| Lev | 0.416** | -0.721 | -0.176*** | 0.418** | -0.830 | -0.177*** |
| | (2.34) | (-0.21) | (-11.42) | (2.35) | (-0.24) | (-11.34) |
| Largest | -0.084 | 20.027*** | -0.054*** | -0.085 | 19.920*** | -0.054*** |
| | (-0.41) | (4.72) | (-3.15) | (-0.41) | (4.67) | (-3.15) |
| Growth | -0.219** | -0.787 | -0.001 | -0.218** | -1.045 | -0.001 |
| | (-2.18) | (-1.17) | (-0.21) | (-2.18) | (-1.54) | (-0.49) |
| PPE | -1.482*** | 39.632*** | -0.082*** | -1.503*** | 40.866*** | -0.077*** |
| | (-3.05) | (9.08) | (-5.01) | (-3.08) | (9.24) | (-4.75) |
| Constant | 2.092** | -73.264*** | 0.196*** | 2.287** | -88.064*** | 0.144*** |
| | (2.41) | (-4.44) | (3.46) | (2.46) | (-5.38) | (2.60) |
| Industry FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 12,227 | 12,170 | 12,251 | 12,227 | 12,170 | 12,251 |
| R-squared | 0.070 | 0.207 | 0.424 | 0.070 | 0.198 | 0.420 |

Notes: This table shows the financial consequences of customer concentration (*CC*). Columns (1) - (2) report the regression results for *CustomerSales*, and columns (3) - (4) report the results for *CustomerHHI*. The dependent variables are *Inventory, ARTurn* and *Profit*. Relevant control variables, as well as industry- and year-fixed effects are included in all regressions. See Table 1 for variable definitions. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 0.01, 0.05, and 0.10 level, respectively.

represents the subsample of firms below the median in the percentage of sales from large buyers. The main results are reported in Table 7. First, the estimated coefficient of inventory on delayed payments is positive and statistically significant for the full sample and HIGH subsample, but it is statistically insignificant for the LOW subsample. This implies that the potential adverse effect of delayed payments on inventory is mainly driven by the delayed payments from large buyers. Next, the results in Table 7 also suggests that delayed payments would reduce a firm's investment and deteriorate its financing constraints. However, it appears that the adverse effects of delayed payments on investment and financing constraints are driven by the delayed payments from both small and large buyers. Taken together, delayed payments from large buyers should have adverse impacts on investment, inventory, and financing constraints.

Furthermore, it is sometimes argued that customer concentration can also adversely impact a firm's creditworthiness. Since the financing constraints of the firm will deteriorate with the worsening creditworthiness, we test this argument by examining the relationship between customer concentration and financing constraints. The results in Table 8 show that financing constraints of firms deteriorate with customer concentration, indicating the adverse impact of customer concentration on a firm's creditworthiness. These results are consistent with the findings of Campello and Gao (2017) and Ni et al. (2023).

## 4. Conclusion

In this paper, we have empirically analyzed the potential passive risk-taking behavior of Chinese firms caused by customer concentration, as motivated by the tentative argument of Campello and Gao (2017) that customer concentration can drive firms to make riskier investments. Using a panel data on Chinese listed companies during 2008–2019, we find that customer concentration does give rise to the passive risk taking of the upstream firms. This effect is economically significant, since a one-standard deviation increase in customer concentration can roughly increase corporate risk-taking by 12% and decrease profit by 5%.

Moreover, we show that a concentrated customer base may drive upstream firms to make risky investments, such as excessive inventory, and weaken the supplier's receivables turnover. The heterogeneity analysis further reveals that the effect of customer concentration on corporate risk-taking is more pronounced for non-SOEs and firms with a relatively weak bargaining position. Through these findings, our analysis shows a type of passive risk-taking behavior induced by customer concentration in supply chain interactions, which indicates a negative side of risk-taking that is largely ignored in the literature on corporate risk-taking.

## CRediT authorship contribution statement

**Xiyang Cao:** Conceptualization, Investigation, Writing – original draft, Visualization. **Jian Ni:** Conceptualization, Writing – review

**Table 7**
The impact of delayed payments from large buyers: sub-sample analyses.

| Panel A: Inventory and investment | | | | | | |
|---|---|---|---|---|---|---|
| Dep.Var: | *Inventory* | | | *Investment* | | |
| Subsample: | FULL | HIGH | LOW | FULL | HIGH | LOW |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Delayed payments* | 0.149*** | 0.212*** | 0.083 | -0.016*** | -0.020** | -0.015* |
| | (3.14) | (3.23) | (1.39) | (-2.81) | (-2.51) | (-1.83) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 12,217 | 6106 | 6110 | 7821 | 3914 | 3906 |
| *R*-squared | 0.556 | 0.404 | 0.651 | 0.167 | 0.165 | 0.182 |
| Panel B: Financing constraints | | | | | | |
| Dep.Var: | *KZ* | | | *OCF* | | |
| Subsample: | FULL | HIGH | LOW | FULL | HIGH | LOW |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Delayed payments* | 1.585*** | 1.304*** | 1.944*** | -0.070*** | -0.060*** | -0.079*** |
| | (15.76) | (10.44) | (12.73) | (-16.44) | (-10.89) | (-12.75) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 12,240 | 6121 | 6118 | 12,240 | 6121 | 6118 |
| *R*-squared | 0.617 | 0.621 | 0.621 | 0.279 | 0.250 | 0.317 |

*Notes:* This table shows the sub-sample analyses regarding the effects of large buyers' delayed payments on investment, inventory, and financing constraints. We obtain regression results on three different samples: FULL, HIGH, and LOW, where FULL represents the full sample, HIGH indicates the subsample of firms above the median in the percentage of sales from large buyers (top five customers), and HIGH represents the subsample of firms below the median in the percentage of sales from large buyers. The dependent variable, *Delayed payments*, is measured by the ratio of receivables to sales. In panel A, the dependent variable is *Inventory (measured by* the ratio of inventory to sales) and *Investment (measured by* the ratio of capital expenditures to lagged assets). In panel B, the dependent variable is *KZ* (the proxy variable constructed as a linear combination of five financial ratios in Lamont et al. (2001)) and *OCF* (the ratio of net operating cash flow to total assets). All regressions include the full set of covariates from the baseline regression, as well as industry- and year-fixed effects. See Table 1 for variable definitions. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 0.01, 0.05, and 0.10 level, respectively.

**Table 8**
Customer concentration and financing constraints.

| | *CustomerSales* | | *CustomerHHI* | |
|---|---|---|---|---|
| Dep. var.: | (1) | (2) | (3) | (4) |
| *Financing constraints* | *KZ* | *OCF* | *KZ* | *OCF* |
| *CustomerConcentration* | 0.374*** | -0.020*** | 0.354** | -0.015** |
| | (4.04) | (-5.50) | (1.98) | (-2.00) |
| Controls | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 12,251 | 12,251 | 12,251 | 12,251 |
| *R*-squared | 0.609 | 0.260 | 0.608 | 0.257 |

*Notes:* This table shows the relation between customer concentration and financing constraints. Columns (1) and (2) report the regression results for *CustomerSales*, and columns (3) and (4) report the results for *CustomerHHI*. The dependent variable, financing constraints, is measured by *KZ* (the proxy variable constructed as a linear combination of five financial ratios in Lamont et al. (2001)) in columns 1 and 3, and *OCF* (the ratio of net operating cash flow to total assets) in columns 2 and 4. All regressions include the full set of (unreported) covariates from the baseline regression, as well as industry- and year-fixed effects. See Table 1 for variable definitions. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 0.01, 0.05, and 0.10 level, respectively.

& editing, Funding acquisition. **Fei Wang:** Investigation. **Yue Xu:** Conceptualization, Validation, Writing – review & editing.

## Data availability

Data will be made available on request.

## Appendix

Tables 1–8

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